

The $\cos 2\phi$ modulation on tensor-polarized ^6Li : projected measurements, the coherent intact-recoil channel, and what stands behind the numbers

2026-08-10 · development run 5 · polligen (evgen/) + polli_fastsim (fastsim/) · companion documents: plans/06 (background budget), docs/note_cos2phi_coherent_6Li.md (physics note) · all external numbers verified against primary sources in three research sweeps; adversarially reviewed

This report presents two projected measurements ("money plots") for a transversely tensor-polarized ^6Li beam at the EIC, explains the design choices behind the simulation machinery and the physics formulas, states every critical assumption explicitly, and lists the improvements that would firm the projections up. The observable is the $\cos 2\phi$ azimuthal modulation driven by photon double-helicity flip — "exotic glue" [1,2] — measured two ways: inclusively (electron only), and in coherent diffraction with the ^6Li detected intact in its 1^+ ground state.

13.5%

coherent-tag acceptance at IP6, high-acceptance optics (exp(-B·p_T²/cut); 9–20% for $B \in 40\text{--}60 \text{ GeV}^2$). High-divergence: 4×10^{-5} — the coherent program fixes the optics choice.

1.1×10^8

tagged coherent events at $100 \text{ fb}^{-1}/\text{u}$ (scenario coherent fraction $f_0 = 0.04$); best bin $\delta A = 6 \times 10^{-4} \rightarrow 5\sigma$ floor at a 0.3% modulation

0.036

deformation-anchored (a_2) of the tagged sample at $P_{zz} = 0.6$ [band 0.018–0.059], scaled from the polarized-deuteron CGC calculation [13] — sign flip vs the deuteron predicted

1 · The observable and why these formulas

For unpolarized electrons on a spin-1 target whose tensor polarization axis lies in the transverse plane, the inclusive cross section is [1,2]:

$$d\sigma/(dx \, dy \, d\phi) = (2y\alpha^2/Q^2) \cdot [F_1 + \frac{1}{2} a_m b_1 + (1-y)/(xy^2) \cdot (F_2 + \frac{1}{2} a_m b_2) - (1-y)/y^2 \cdot c_m \sin^2\theta_m \cdot \Delta(x, Q^2) \cdot \cos 2\phi]$$

$$a_m = \frac{1}{2} c_m (3\cos^2\theta_m - 1), \quad c_m = 3|\lambda_m| - 2 \rightarrow (1, -2, 1)$$

Why this observable. The $\cos 2\phi$ term requires the virtual photon to flip helicity by two units. A spin-1/2 nucleon cannot absorb two units of helicity, so no quark — and no bound nucleon — contributes at leading twist: $\Delta(x, Q^2)$ is purely gluonic and vanishes for a nucleus that is "just protons and neutrons" [1]. It has never been measured for any target; the only prior experimental document is a JLab letter of intent with no sensitivity projection [3]. Magnitude guidance spans per-mille (Sather–Schmidt bag moment $-0.012 \alpha_s$ [4]) to few-percent (maximal $\Delta_T g = \Delta g$, explicitly an overestimate [5]) — hence the scenario band $\Delta/F_1 \in \{10^{-3} \dots 10^{-2}\}$.

Formula choices made and their cost. The master formula is implemented in the Hoodbhoy–Jaffe–Manohar normalization [2] with the fastsim approximation set: $A_{//} = D \cdot g_1/F_1$ (γ^2 -suppressed η term dropped), $g_2 = g_2^{WW}$, and no target-mass/ γ^2 corrections. The last is a real limitation only at high x and low Q^2 ; the sweet-spot bins sit at $x \leq 0.06$ where $\gamma^2 = 4M^2x^2/Q^2 < 10^{-2}$ even at $Q^2 \approx 1 \text{ GeV}^2$, so it is a labeling caveat, not a bias, for these plots. Structure functions are TOY parameterizations with PDF-grid (CT18NLO/NNPDFpol11) backends behind the same interface — rates are trustworthy to a factor ~ 1.5 , asymmetry *shapes* are scenarios by construction.

2 · Simulation design choices

Why an event-level machinery at all. No public generator produces polarized e+A events for $A > 1$ (verified survey; BeAGLE is unpolarized, DJANGO/PEPSI are nucleon-level longitudinal-only). polligen implements only the missing wheel — the spin-density matrix $\otimes$ structure-function master formula and spin-correlated

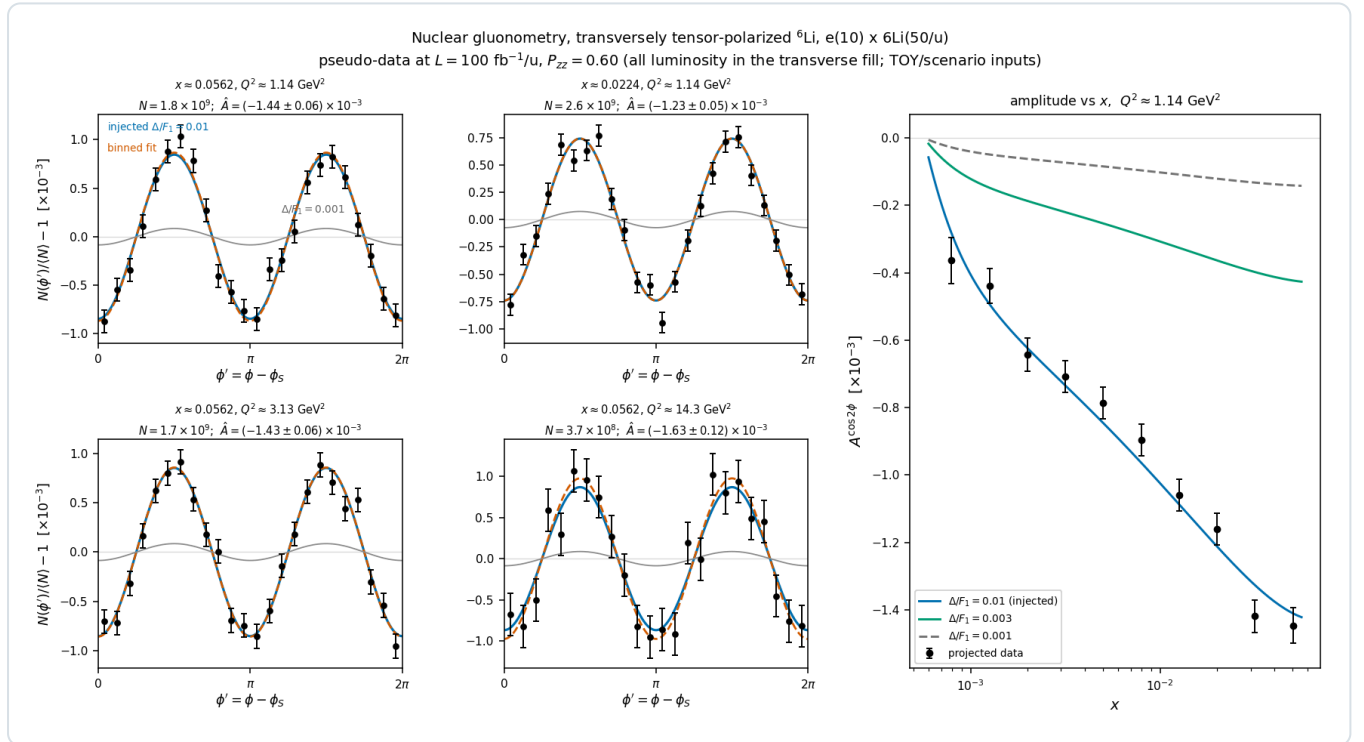
sampling — and reuses everything else. Its Step-5.A validation gates: bin-wise identity with the analytic asymmetry formulas at $\text{rtol } 10^{-12}$, pseudo-experiment error spreads closing on every analytic FOM map, and unbiased $\cos 2\phi$ recovery with holey ϕ acceptance.

Full-luminosity statistics without 10^9 -event files. The money plots need pseudo-data at N up to $\sim 10^9$ events per bin. For a *binned* estimator, drawing each ϕ -bin's count from a Poisson with the exact expected yield (the analytic integral of the modulated cross section over the bin) is statistically identical to event-level sampling — so a full-luminosity pseudo-experiment costs 24 Poisson draws (`sample.phi_histogram_pseudo`). Event-level sampling remains available and is what the closure tests use.

Estimator choice. The amplitude is extracted by a binned least-squares fit of $N(\phi') = C(1 + A \cos 2\phi')$ (`estimators.cos2phi_fit_binned`), not the naive $2\langle \cos 2\phi' \rangle$ moment: with realistic ϕ -acceptance holes the moment is demonstrably biased by more than the signal, while the fit — with bins probed across their full width and dropped if they straddle a hole — stays unbiased. Statistical errors follow $\delta A = \sqrt{(2/N)/P_{zz}}$, validated by pseudo-experiment spreads. A precondition is documented: the fit model carries no $\cos \phi'$ term; the m-symmetric tensor fills guarantee $a_1 = 0$ identically by design.

Sweet-spot selection. The four displayed (x, Q^2) super-bins are chosen from the analytic significance map *per Q^2 band* rather than by raw ranking — the raw map clusters at the lowest Q^2 (rate-dominated), but the Q^2 lever arm is itself physics: it is the consistency check against power-suppressed non-partonic contributions (§5). Each super-bin merges 3×3 cells of the analysis grid (~ 2 in x and Q^2).

Rate conventions. Per-nucleon luminosity and F_2A/A structure functions throughout; scattered-electron acceptance cuts ($|\eta| \leq 3.5$, $E' \geq 0.5$ GeV, $0.01 \leq y \leq 0.95$, $W^2 \geq 10$ GeV²) applied in the rate maps; the whole luminosity is assigned to the transverse-tensor fill — the same convention as the analytic FOM maps (a run-plan split would scale N down proportionally).



Money plot 5 — inclusive gluonometry ($e10 \times {}^6\text{Li}50$, $L = 100 \text{ fb}^{-1}/u$, $P_{zz} = 0.60$, generated at $\Delta/F_1 = 10^{-2}$). Left 2×2 : ϕ' -modulation pseudo-data with statistical error bars in the four sweet-spot super-bins ($Q^2 = 1.1 \rightarrow 14 \text{ GeV}^2$, $x \approx 0.02\text{--}0.06$); injected curve (blue), binned fit (vermillion), $\Delta/F_1 = 10^{-3}$ scenario (gray). Right: extracted amplitude vs x along the best Q^2 slice with three scenario curves. Per-bin $\delta A = (0.4\text{--}1.2) \times 10^{-4}$; at $\Delta/F_1 = 10^{-3}$ this is $2.5\text{--}2.7\sigma$ per low/mid- Q^2 bin, 4.6σ for the four bins shown, 9.7σ combining the full accepted map (numbers independently reproduced by the adversarial review). Detection needs only the scattered electron.

3 · The coherent channel: model choices and the anchored amplitude

Tagging. The intact ${}^6\text{Li}$ recoil has $A/Z = 2$ — exactly beam rigidity ($R = 1.000$; the diffractive momentum loss $x_P < 0.01$ changes rigidity by $< 1\%$). At IP6 it is visible *only* in the Roman-Pot near-beam p_T tail above the 100 beam-envelope cut [7,9]. With an exponential coherent t-distribution the acceptance is analytic, $\exp(-B \cdot p_T^2, \text{cut})$: **13.5%** for the documented high-acceptance cut ($p_T > 0.20$ GeV), **4×10^{-5}** for high-divergence (0.45 GeV — a derived, not documented, number) — the coherent program fixes the optics choice. The IR-8 secondary focus reaches $p_T \approx 0$; published intact-recoil efficiencies (d 47%, ${}^3\text{He}$ 32%, ${}^4\text{He}$ 29%, ${}^7\text{Li}$ 17.8% [10]) interpolate to $\approx 20\%$ for ${}^6\text{Li}$ — an independent cross-check.

Rate model choices (`polligen/coherent.py`, all bands explicit): $\sigma_{\text{coh}} = f_{\text{coh}}(x) \cdot \sigma_{\text{DIS}}$ with $f_{\text{coh}} = f_0 / (1 + (x/x_{\text{coh}})^2)$, $f_0 = 0.04$ [0.02–0.08] — no light-nucleus prediction exists (lightest published: Ca), so the band brackets HERA ep diffraction (10–15% of DIS [44]) and heavy-A saturation estimates (20–25% [45]); $x_{\text{coh}} = 0.01$ from the coherence length $l_c = 0.105 \text{ fm}/x_P$ vs the ${}^6\text{Li}$ size; slope $B = 50 \text{ GeV}^{-2}$ [40–60] from $B = \langle r^2 \rangle / 3$ with the matter radius (charge radius gives ≈ 57 [47]; the gluonic B may sit higher [42]), exponential valid below the form-factor minimum at $|t| \approx 0.31 \text{ GeV}^2$ [11,12]. Writing f_{coh} against Bjorken x (not x_P) is optimistic for high-mass diffraction; the f_0 band covers it.

The amplitude is anchored, not guessed. The polarized-deuteron CGC calculation of Mäntysaari, Salazar, Schenke, Shen, and Zhao [13] is the *only* published calculation of polarization-dependent azimuthal coefficients in coherent diffraction on any polarized nucleus (all 8 forward citations checked; the group's follow-ups are unpolarized [14,15]). Their figures, digitized from the arXiv source into `coherent.MANTYSAARI_A2_DEUTERON`: $a_2(m=0) = +0.08/+0.15/+0.30/+0.43$ at $|t| = 0.05/0.1/0.2/0.3 \text{ GeV}^2$, $a_2(m=\pm 1) \approx -a_2(m=0)/2$, a factor-2 m-state cross-section difference at $|t| = 0.1$ — all from the 6% D-wave modulating the diffractive slope, $a_2 \approx (\Delta B_m/2) \cdot |t|$ with $\Delta B_0/B \approx 0.21$. For a fill with populations p_m the ensemble coefficient is exactly $\propto P_{zz}$:

$$a_2(t; P_{zz}) = -(P_{zz}/4) \cdot \epsilon_{B0} \cdot B \cdot |t|, \quad \epsilon_{B0} \equiv \Delta B_0/B$$

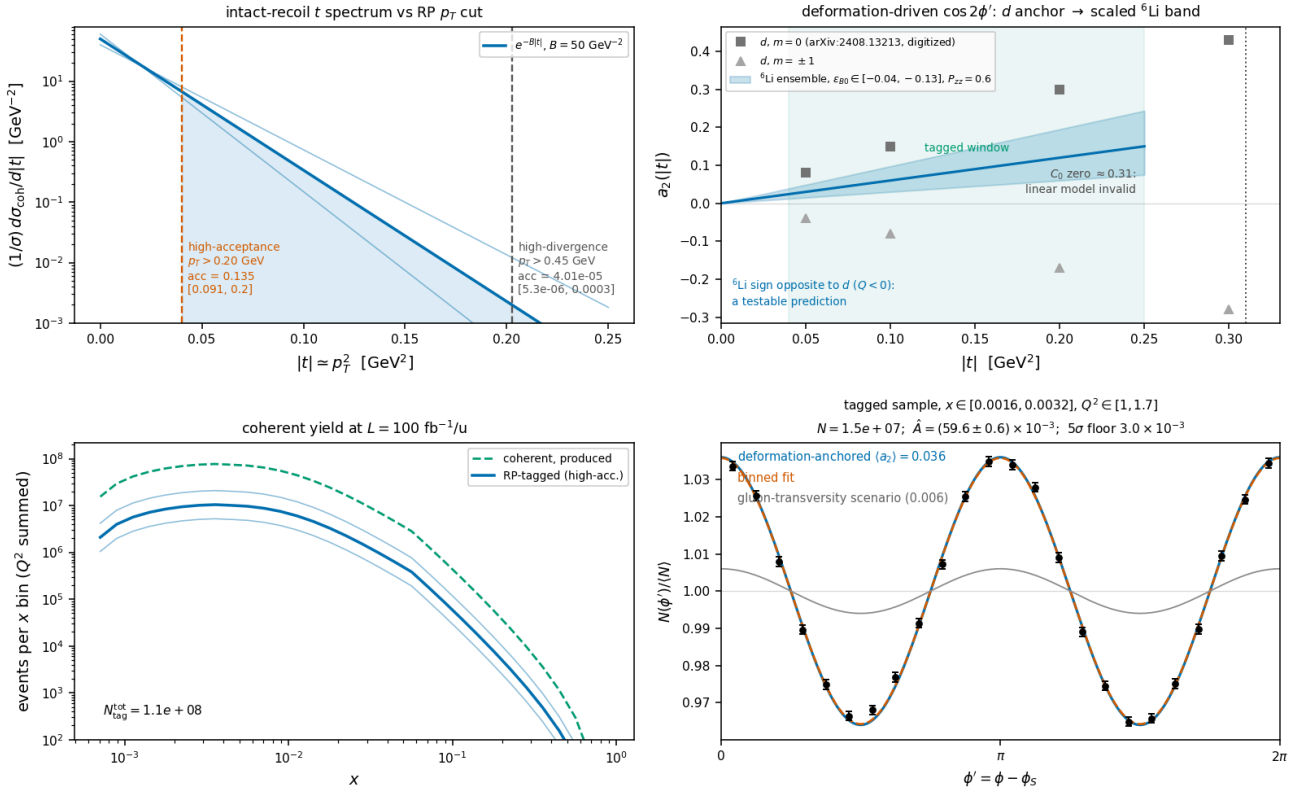
${}^6\text{Li}$ scaling of ϵ_{B0} (two dials that usefully disagree):

- relative quadrupole deformation $Q/\langle r^2 \rangle$: ${}^6\text{Li} \approx (1/5) \times \text{deuteron}$, OPPOSITE sign
- α -d asymptotic D/S ratio $\eta = -0.010 \dots -0.015$ [16] vs $\eta_d = +0.0256$: $0.4\text{--}0.6 \times$
 $\Rightarrow \epsilon_{B0}({}^6\text{Li}) \in -(0.04 \dots 0.13)$, default -0.08
 $\Rightarrow \langle a_2 \rangle_{\text{tag}} = a_2(\langle |t| \rangle_{\text{tag}} = p_T^2, \text{cut} + 1/B) \approx 0.036$ [0.018–0.059] at $P_{zz} = 0.6$

The **sign flip relative to the deuteron is a testable prediction**. The band is physics, not sloppiness: even exact few-body theory misses the ${}^6\text{Li}$ quadrupole cancellation by $\sim 3 \times$ [17], and the gluonic deformation need not cancel like the charge one. Existence proofs at exactly this $|t|$: measured e-d $T_{20} = -(0.1\text{--}0.3)$ [19]; HERMES $A_{zz}(d) \approx -1\%$ at $x \approx 0.01$ [20], coherent-double-scattering in origin [21–23]. The flat gluon-transversity term is separately bounded at $3 \times 10^{-3}\text{--}10^{-2}$ (lattice ϕ -meson transversity [25,26] $\div$ the $\sim 10 \times$ NPLQCD nuclear suppression [27]; Kumano–Song few-% only at maximal Δ_T g [5,28]).

The measurement this buys: the two mechanisms separate by shape — deformation is t-linear, sign-flipped vs the deuteron, and should weaken toward small x_P (JIMWLK washes out deformation [15]); exotic glue is flat in t and survives at small x. A two-component fit $a_2(t) = c_{\text{def}} |t| + a_g$ over the tagged window, per x_P bin, with both components $\geq 5\sigma$ above the statistical floor at $100 \text{ fb}^{-1}/u$. Incoherent breakup is m-state-blind in this mechanism [13] — it dilutes but cannot fake the modulation.

Coherent $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$, transversely tensor-polarized, $e(10) \times 6\text{Li}(50/u)$, $P_{zz} = 0.60$
(scenario rates; modulation anchored on the polarized- d CGC calculation, PLB 858:139053 -- plans/06 SS6.4b)



Money plot 6 — coherent channel, $e^6\text{Li} \rightarrow e'X^6\text{Li(g.s.)}$ (same beam, $P_{zz} = 0.60$). Top left: recoil t -spectrum vs both optics' p_T cuts with the slope band. Top right: the amplitude anchor — digitized deuteron a_2 per m -state [13] vs the scaled ^6Li ensemble band (sign flip predicted); tagged window and C_0 -zero exclusion marked. Bottom left: tagged yield vs x at $100 \text{ fb}^{-1}/u$ with the f_0 band ($N_{\text{tag}} \approx 1.1 \times 10^8$). Bottom right: ϕ' pseudo-data at the deformation-anchored $\langle a_2 \rangle = 0.036$ with the flat gluon-transversity scenario (grey); best-bin extraction $\hat{A} \pm 0.6 \times 10^{-3}$.

4 • Backgrounds for the coherent tag (summary — full budget in plans/06)

BACKGROUND	WHY IT FAKES THE TAG	HANDLE
$\alpha + d$ breakup (threshold 1.474 MeV; 2.186 MeV 3^+ , $\Gamma = 24$ keV, $\sim 100\%$ $\alpha + d$ [39])	Both fragments at $R = 1.000$, same trajectory <i>and</i> velocity as intact ^6Li — tracking and ToF blind; only dE/dx ($\propto Z^2$: 9/4/1) separates	RP charge discrimination — open question #19 : no EIC document addresses the $A/Z = 2$ degeneracy (AC-LGAD amplitude exists but is used for position; documented Z-ID is an IR-8 Cherenkov [8]). Fallback: two-component $ t $ fit (coherent $e^{-50 t }$ vs $\sim 10\times$ flatter incoherent; crossover at $0.05\text{--}0.2 \text{ GeV}^2$ sits inside the window [41,42]); $e + \text{Pb}$ benchmark: 80–99% incoherent rejection [43]
Nucleon-emission breakup: $\alpha + p + n$ (3.70 MeV), $^3\text{He} + t$ (15.79 MeV) [39]	A fragment could pass unnoticed...	...but $p \rightarrow \text{OMD}$ ($R = 0.50$), $n \rightarrow \text{ZDC}$, $^3\text{He} \rightarrow \text{RP}$ main window ($R = 0.75$): vetoable; inefficiency is second-order (BeAGLE study to quantify)
$^6\text{Li}^*$ 3.5629 MeV (0^+ , $T = 1$, $\Gamma = 8.2 \text{ eV}$) [39]	Particle-stable ($\alpha + d$ strictly parity-forbidden, $\Gamma_\alpha \leq 6 \times 10^{-7}$ eV); γ -decays $\rightarrow$ real $A=6$, $Z=3$ track	Isoscalar exchange cannot drive $T=0 \rightarrow T=1$; the boosted $0.1\text{--}0.4 \text{ GeV}$ γ lands mostly in the B0 EMCAL ($\geq 50\text{--}100 \text{ MeV}$, $5.5\text{--}20 \text{ mrad}$ [9]), tail in ZDC. The 5.366 MeV $2^+ T=1$ always emits a nucleon $\rightarrow$ vetoable. Every $T=1$ state is γ- or nucleon-vetoable; only $T=0$ states feed the blind $\alpha + d$ channel
Bethe-Heitler / QED Compton	Intact nucleus by construction	Calculable (t -dependence = charge FF^2 [11,12]); subtract + reuse as RP- acceptance candle; charge-quadrupole $\cos 2\phi < 10^{-3}$ for $ t < 0.15 \text{ GeV}^2$ (^6Li C2 unobservably small below $q \approx 3 \text{ fm}^{-1}$ [12,17])
Beam-gas / halo; e' -side fakes	Accidental $R \approx 1$ tracks; π /pair- symmetric e' at high y	Non-colliding bunches, $\sim 30 \text{ ps}$ timing, vertex association; standard DIS e' ID + (e' , recoil) kinematic consistency

Extraction systematics: the modulation is self-normalized within a single fill — relative-luminosity systematics drop out; ϕ -acceptance holes are handled by the binned fit (demonstrated); polarimetry scale $\delta P_{zz}/P_{zz} \approx 3\%$

is common with A_{zz} ; tensor-observable radiative corrections are uncharted and flagged on every plot. Interpretation-level: a $\Delta\Delta$ -isobar admixture can mimic Δ at leading twist [34]; conventional double-scattering generates the low- x b_1 [21–23] — the Q^2 lever arm and the ${}^6\text{Li}/d$ comparison are the discriminants.

The ${}^6\text{Li}$ null test: deformation-driven coherent modulations — the polarized-deuteron imaging signal [13] — are bounded on ${}^6\text{Li}$ by its anomalously small quadrupole moment, $Q({}^6\text{Li}) = -0.0806(6) \text{ fm}^2 \approx Q({}^7\text{Li})/50$ [35]. A sizable flat-in- $t \cos 2\phi$ on ${}^6\text{Li}$ cannot be a shape effect: the mechanisms entangled for the deuteron separate cleanly in the ${}^6\text{Li}/d$ comparison.

5 • Critical assumptions, honestly ranked

#	ASSUMPTION	IMPACT IF WRONG	REPLACEMENT PATH
1	Scenario coherent fraction $f_0 = 0.04$ [$\times 2/\div 2$] and its Bjorken- x (not x_P) parameterization	All coherent yields scale linearly; δA scales as $1/\sqrt{f_0}$	Real diffractive model — open question #18; the concrete ask is an IP-Glasma α - d run [13]
2	ε_{B0} scaling from the deuteron (two dials disagreeing by $\times 3$; charge-quadrupole cancellation may not transfer to gluons)	Deformation amplitude anywhere in the stated band; sign flip robust	Same IP-Glasma α - d calculation; meanwhile the band is quoted on every plot
3	Event-by-event Z-ID in the RPs is NOT assumed (purity via $ t $ -shape fit)	If Z-ID exists, purity improves outright; if the incoherent slope model is off, the fit systematic grows	#19 answer from ePIC FF WG; BeAGLE Mode-W incoherent spectra
4	Linear $a_2(t)$ valid only for $ t < 0.2 \text{ GeV}^2$; C_0 zero at 0.31 GeV^2 excluded, not modeled	Upper tagged window carries unmodeled structure (locally enhanced, sign-flipping a_2)	Form-factor-level model of the dip region (the deuteron analog is in [13])
5	TOY structure functions; no target-mass terms; $g_2 = g_2^{WW}$	Rates to $\times 1.5$; shapes are scenarios by construction	PDF-grid backends exist; digitized theory curves scheduled (plans/02 step 1.2)
6	High-divergence p_T cut 0.45 GeV is derived from YR divergence tables, not documented	Only affects the (already excluded) high-divergence option	ePIC optics documents as they firm up
7	All luminosity in the transverse-tensor fill; $P_{zz} = 0.6$ in-ring placeholder	N scales with the run-plan share; amplitude scales with P_{zz}	Run-plan bookkeeper already supports splits; EPIOS/source numbers as they land (ABS sources: $\sim 95\%$ of maximal P_{zz} demonstrated [37])
8	Impulse approximation; no FSI; tensor radiative corrections unknown	FSI distorts the recoil spectrum at larger $ t $; RC unquantified for tensor observables	Theory engagement (plans/04 #10, #16); QED tail is calculable and doubles as calibration
9	Deformation term assumed x_P -independent within the plotted bins	JIMWLK evolution weakens deformation toward small x [15] — would mimic a relative growth of the gluonic term	Make the x_P dependence part of the two-component fit; theory input from [15]

6 • Future improvements

Near term (this program): BeAGLE Mode-W incoherent ${}^6\text{Li}$ spectra $\rightarrow$ veto-efficiency and $|t|$ -fit systematics become numbers (plans/02 step 1.5); VMC α - d/α - t overlaps replace the two-parameter radial forms (plans/04 #15); pseudo-experiment closure with a 2-D (ϕ , ϕ_t) acceptance map (Phase-2, plans/03 step 2.3); HepMC3 output + ePIC chain smoke test (plans/05 step 5.D) so these projections can be repeated at reconstructed level.

Theory engagement: #18 — rerun the IP-Glasma polarized-deuteron setup [13] with an α - d cluster density for ${}^6\text{Li}$ (the single highest-leverage input: it would replace both ε_{B0} dials and the f_0 band at once); the spin- $3/2$ (${}^7\text{Li}$) structure-function basis (plans/04 #14); tensor-observable radiative corrections (#10); the sum-rule tie between the transversity-GPD moments and the measured tensor form factors [30] to implement the cancellation scenario rigorously.

Detector/machine questions raised by this study: #19 — can the RP AC-LGAD amplitude separate $Z = 3/2/1$ at the same rigidity and velocity? (No EIC document addresses it; decides whether coherent-tag purity is event-by-event or statistical.) The IR-8 secondary focus with its Z^2 Cherenkov [8] is the structural fix and lifts the tag acceptance from 13.5% toward $p_T \approx 0$ coverage (^7Li : 17.8% [10]) — the coherent- ^6Li program is one more physics case for the second IR. Off-momentum halo determines how close the pots sit, i.e. the p_T cut itself.

References

- [1] R. L. Jaffe, A. Manohar, *Nuclear gluonometry*, Phys. Lett. B 223 (1989) 218.
- [2] P. Hoodbhoy, R. L. Jaffe, A. Manohar, *Novel effects in deep inelastic scattering from spin-one hadrons*, Nucl. Phys. B 312 (1989) 571.
- [3] J. Maxwell et al., *Search for Exotic Gluonic States in the Nucleus*, JLab LO12-16-006, arXiv:1803.11206.
- [4] E. Sather, C. Schmidt, *Size and scaling of the double-helicity-flip hadronic structure function*, Phys. Rev. D 42 (1990) 1424.
- [5] S. Kumano, Q.-T. Song, *Gluon transversity in polarized proton-deuteron Drell-Yan process*, Phys. Rev. D 101 (2020) 054011, arXiv:1910.12523.
- [6] Y. Hatta, B.-W. Xiao, F. Yuan, Phys. Rev. Lett. 116 (2016) 202301.
- [7] R. Abdul Khalek et al., *EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419.
- [8] Second EIC detector / IR-8 studies: arXiv:2211.15746 (§VIII); BNL LDRD 23-050 closeout, arXiv:2602.04636.
- [9] M. Pitt, PoS DIS2024 (2025) 259, arXiv:2409.02811; E.-C. Aschenauer et al., Phys. Rev. D 112 (2025) 036010, arXiv:2503.05908; M. Boër et al., arXiv:2512.15064.
- [10] W. Chang, E.-C. Aschenauer, A. Jentsch, A. Kumar, Z. Tu, Z. Yin, *Opportunities for imaging light nuclei with a second interaction region at the EIC*, Phys. Rev. D 113 (2026) 032018, arXiv:2511.05638.
- [11] L. R. Suelzle, M. R. Yearian, H. Crannell, Phys. Rev. 162 (1967) 992.
- [12] G. C. Li, I. Sick, R. R. Whitney, M. R. Yearian, Nucl. Phys. A 162 (1971) 583.
- [13] H. Mäntysaari, F. Salazar, B. Schenke, C. Shen, W. Zhao, *Spatial imaging of polarized deuterons at the Electron-Ion Collider*, Phys. Lett. B 858 (2024) 139053, arXiv:2408.13213.
- [14] H. Mäntysaari, H. Roch, B. Schenke, C. Shen, W. Zhao, Phys. Rev. D 114 (2026) 014068, arXiv:2605.00454.
- [15] H. Mäntysaari, P. Singh, *Energy dependence of the deformed nuclear structure at small- x* , Eur. Phys. J. C 85 (2025) 1449, arXiv:2411.14934.
- [16] K. D. Veal, C. R. Brune, W. H. Geist et al., Phys. Rev. C 60 (1999) 064003.
- [17] R. B. Wiringa, R. Schiavilla, Phys. Rev. Lett. 81 (1998) 4317, nucl-th/9807037.
- [18] M. Garçon, J. W. Van Orden, Adv. Nucl. Phys. 26 (2001) 293, nucl-th/0102049.
- [19] M. Bouwhuis et al. (NIKHEF), Phys. Rev. Lett. 82 (1999) 3755; D. Abbott et al. (JLab t20), Phys. Rev. Lett. 84 (2000) 5053; D. M. Nikolenko et al. (VEPP-3), Phys. Rev. Lett. 90 (2003) 072501.
- [20] A. Airapetian et al. (HERMES), Phys. Rev. Lett. 95 (2005) 242001, hep-ex/0506018.
- [21] N. N. Nikolaev, W. Schäfer, Phys. Lett. B 398 (1997) 245.
- [22] K. Bora, R. L. Jaffe, Phys. Rev. D 57 (1998) 6906, hep-ph/9711323.
- [23] J. Edelmann, G. Piller, W. Weise, Phys. Rev. C 57 (1998) 3392.
- [24] A. Airapetian et al. (HERMES), Nucl. Phys. B 842 (2011) 265, arXiv:1008.3996.
- [25] W. Detmold, P. E. Shanahan, Phys. Rev. D 94 (2016) 014507, arXiv:1606.04505.
- [26] W. Detmold, P. E. Shanahan, Phys. Rev. D 95 (2017) 114515, arXiv:1703.08220.
- [27] F. Winter et al. (NPLQCD), Phys. Rev. D 96 (2017) 094512, arXiv:1709.00395.
- [28] S. Kumano, Q.-T. Song, Phys. Rev. D 101 (2020) 094013, arXiv:2003.06623.
- [29] J. Zhao, A. Bacchetta, S. Kumano, T. Liu, Y.-J. Zhou, JHEP 12 (2025) 067, arXiv:2508.06134.
- [30] W. Cosyn, B. Pire, Phys. Rev. D 98 (2018) 074020, arXiv:1806.01177; W. Cosyn et al., Phys. Rev. D 99 (2019) 094035; W. Cosyn, B. Pire, L. Szymanowski, PoS DIS2019 (2019) 254, arXiv:1907.08662.
- [31] D. Boer, S. Cotogno, T. van Daal, P. J. Mulders, A. Signori, Y.-J. Zhou, JHEP 10 (2016) 013, arXiv:1607.01654; S. Cotogno, T. van Daal, P. J. Mulders, JHEP 11 (2017) 185, arXiv:1709.07827.
- [32] E. R. Berger, F. Cano, M. Diehl, B. Pire, Phys. Rev. Lett. 87 (2001) 142302.
- [33] F. Cano, B. Pire, Eur. Phys. J. A 19 (2004) 423, hep-ph/0307231; A. Kirchner, D. Müller, Eur. Phys. J. C 32 (2003) 347, hep-ph/0302007.
- [34] M. Nzar, P. Hoodbhoy, Phys. Rev. D 45 (1992) 2264.
- [35] I. Angeli, K. P. Marinova, At. Data Nucl. Data Tables 99 (2013) 69; Q values via arXiv:2403.06384; N. J. Stone, At. Data Nucl. Data Tables 111–112 (2016) 1.
- [36] B. Wojtsekhowski, arXiv:2406.11480.
- [37] M. M. Dalton, A. Deur, C. D. Keith, Eur. Phys. J. A 61 (2025) 111, arXiv:2504.21177.
- [38] M. M. Dalton, A. Deur, C. D. Keith, N. Fomin, M. Sargsian, JLab PAC 53 proposal, arXiv:2508.06481.
- [39] D. R. Tilley et al., *Energy levels of light nuclei $A = 5, 6$* , Nucl. Phys. A 708 (2002) 3 (TUNL).
- [40] arXiv:2211.15746 §VIII; arXiv:2602.04636 (IR-8 Z^2 Cherenkov).
- [41] LEPS collaboration, coherent ϕ on ^4He , arXiv:1711.01095.
- [42] STAR collaboration, Phys. Rev. Lett. 128 (2022) 122303, arXiv:2109.07625.
- [43] W. Chang et al., Phys. Rev. D 104 (2021) 114030, arXiv:2108.01694.
- [44] H1 collaboration, Eur. Phys. J. C 72 (2012) 2074, arXiv:1203.4495.
- [45] T. Lappi, Nucl. Phys. A 830 (2009) 403c, arXiv:0907.4588; H. Kowalski, T. Lappi, C. Marquet, R. Venugopalan, Phys. Rev. C 78 (2008) 045201, arXiv:0805.4071.
- [46] A. Accardi et al., Eur. Phys. J. A 52 (2016) 268, arXiv:1212.1701.
- [47] I. Sick, arXiv:1505.06924.
- [48] ECCE consortium, arXiv:2208.14575.
- [49] Polarized $^3\text{He}/^4\text{He}$ coherent DVCS at the EIC, arXiv:2606.11491.
- [50] G. Atoian et al. (EPIOS), Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794.

Generated from the working tree of the PolarizedLithiumSim repository (development run 5, 2026-08-10): `evgen/scripts/money_cos2phi.py`, `evgen/scripts/money_cos2phi_coherent.py`, model in `evgen/polligen/coherent.py`; 66 tests pass. All rates use TOY/scenario inputs — bands, not predictions. External numbers verified against primary sources in three research sweeps (nuclear data: TUNL; detector: EIC YR + ePIC documents; theory: original papers); citation metadata individually verified against arXiv/INSPIRE. Target venue: INT program on polarized ion beams at the EIC, March 22 – April 2, 2027.